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Applying a novel two-stage deep-learning model to improve accuracy in detecting retinal fundus images

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ABSTRACT

Applications of artificial intelligence (AI) in medical imaging informatics have attracted broad research interest. In ophthalmology, for example, automated analysis of retinal fundus photography helps diagnose and monitor illnesses like glaucoma, diabetic retinopathy, hypertensive retinopathy, and cancer. However, building a robust AI model requires a large and diverse dataset for training and validation. While large number of fundus photos are available online, collecting them to create a clean, well-structured dataset is a difficult and manually intensive process. In this work, we propose a two-stage deep-learning system to automatically identify clean retinal fundus images and delete images with severe artifacts. In two stages, two transfer-learning models based the ResNet-50 architecture pre-trained using ImageNet data are built with Increased threshold values on SoftMax to reduce false positives. The first stage classifier identifies “easy” images, and the remaining “difficult” (or undetermined) images are further identified by the second stage classifier. Using the Google Search Engine, we initially retrieve 1,227 retinal fundus images. Using this two-stage deep-learning model yields a positive predictive value (PPV) of 98.56% for the target class compared to a single-stage model with a PPV of 95.74%. The two-stage model helps reduce by two-thirds the false positives for the retinal fundus image class. The PPV over all classes increases from 91.9% to 96.6% without compromising the number of images classified by the model. The superior performance of this two-stage model indicates that the building of an optimal training dataset can play an important role in increasing performance of deep-learning models.

Keywords: Deep Learning, Retinal Fundus, Transfer Learning, Multi-Stage Classification, Retinal Photography, Positive Predictive Value, K-fold Cross Validation

1. INTRODUCTION

In medical image analysis, clinical data representation and extraction is the primary purpose, and the premise of many complicated frameworks. Many voluntary and involuntary models have been developed in the field [1]–[3]. These models assist clinicians or researchers to simplify and evaluate medical images in various applications, e.g., organ segmentation [4]–[6], anatomical landmark detection [7], [8], and computer aided diagnosis (CAD) [9]–[12]. Various organ systems have highly diverse attributes and the medical image models are usually designed for specific studies to assimilate appropriate prior knowledge. However, models using handcrafted features or parameter settings are often difficult to optimize and not easily adaptable to develop other CAD schemes and applications.

Thus, in the last decade, deep-learning technology and models [13] have emerged and attracted extensive research and development interest in broad scientific communities. Deep-learning models have performed successfully in various applications, including natural-language processing [14], signal processing [15], object detection [16]. Many variants of deep learning architectures addressing the diabetic retinopathy were also proposed [17]. Their popularity comes from their ability to model nonlinear and hierarchical features from many complex datasets. Deep network architectures are able to extract features under supervised or unsupervised environment from large quantities of training data. Image recognition [16] is a promising application where deep learning learns good visual features. Likewise, deep learning may be put to good use for medical image recognition as well.

A retinal fundus photograph consists of a color image of an eye's interior surface. Such photographs help ophthalmologists identify disorders. Examining retinal fundus (ocular fundus) imagery has become increasingly valuable to other medical fields besides ophthalmology. Neurologic disorder early indications can be found using optic nerve, retina, and choroid abnormalities [18]. Such imagery also helps neurologists understand underlying systemic diseases that contribute to a patient's neurological condition. Hence, fundus photography can help physicians to identify severe conditions. Developing computed aided diagnosis (CAD) models using artificial intelligence (AI) or deep machine learning (ML) with superior performance to identify such underlying conditions requires a large, clean, and diverse dataset. Due to HIPAA and other regulations, these datasets can be difficult to acquire from private sources.

Using public data platforms such as the Google search engine is one method to obtain such a dataset. However, when building the dataset using appropriate keywords in the search engine, many non-useful and noisy images with severe artifacts or overlapped un-related items are also returned along with the useful fundus photography images. To select useful and clean fundus images visually or manually is a tedious and time-consuming task.

In order to improve efficacy of training and applying deep learning models using relatively small image datasets, the objective of this study is to investigate the feasibility and advantages of developing a new automated scheme to identify and select useful retinal fundus images from all downloaded images without human intervention. Each downloaded image is to be categorized into one of three classes based on its content: Class 1 – complete, single retinal fundus images without annotations; Class 2 – single images containing multiple fundus photos or single fundus photos with annotations; Class 3 – noisy images overlapped with artifacts. Class 1 is the target class that goes into the database. In this study, it is essential to achieve the highest possible PPV for Class 1 images.

2. MATERIALS AND METHODS

2.1. Dataset

We use the Google search engine to query and retrieve retinal fundus photos or images. A total of 1,227 unique images are downloaded from the Google search engine. These images are downloaded in batches using different keywords related to retinal fundus images. Each image is classified into one of three classes. Each image in Class 1 is a single fundus photo without any annotations as shown in Figure 1. Each image in Class 2 is a single fundus photo with annotations or multiple fundus photos with at least 25% of the total area of the image containing a fundus photo as shown in Figure 2. Class 3 consists of images in which the fundus photos comprise less than 25% total area of the image and other noisy images as shown in Figure 3. Based on above criteria, Class 1 has 620 images, Class 2 has 134 images, and Class 3 has 473 images.

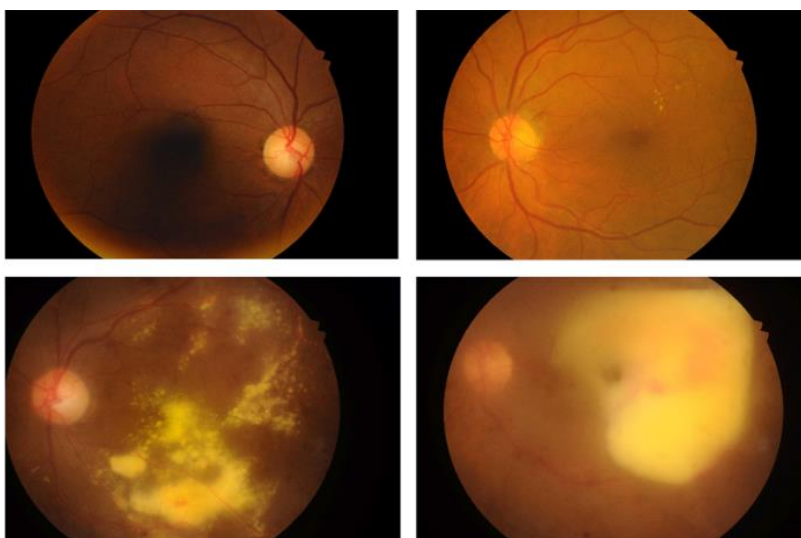


Figure 1: Four example images of Class 1

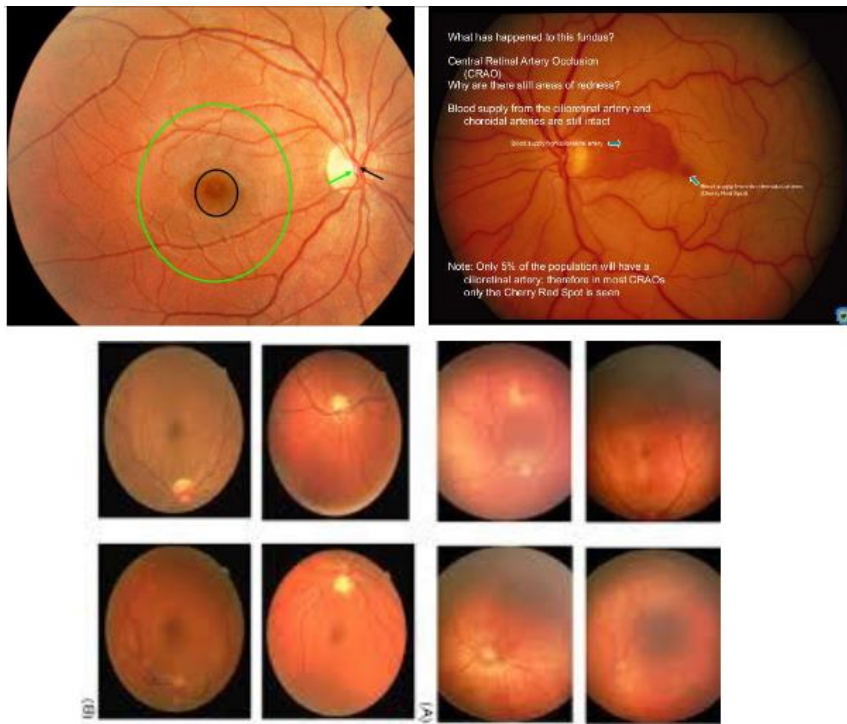


Figure 2: A sample of Class 2 images. Two on the top and one on the bottom.



Figure 3: A sample of Class 3 images.

2.2. Methods

We use transfer learning to build new deep-learning models to classify the retinal fundus images into 3 classes. For this purpose, the ResNet-50 [19] architecture is selected to build our transfer learning model, which has been pre-trained using a large ImageNet database. Although many Deep Convolutional Neural Network (DCNN) models have been developed and used as transfer learning models in medical imaging informatics field, we chose a DCNN model using ResNet-50 architecture. ResNet-50 has several advantages compared with VGG19 [20] such as a smaller number of parameters to train (more depth, less width), reduced effect of vanishing gradient, and higher accuracy obtained on the ImageNet dataset.

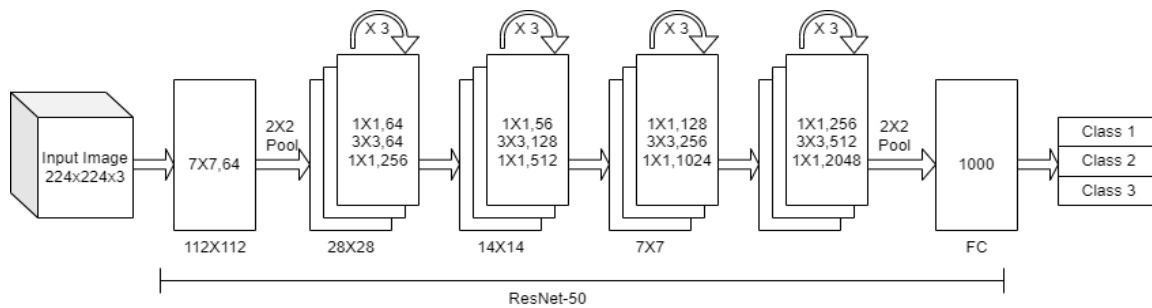


Figure 4: Block diagram of stage1 transfer learning with ResNet-50

In our study, we first randomly select 50 images from each of the three classes to build a training dataset. Thus, a total of 150 images are used to fine-tune the ResNet-50 model. The fine-tuned ResNet-50 model is then applied to the remaining 1077 images to classify these images into three classes. Since the model generates three probability scores in three classes, the image is assigned to the class that has the highest probability score.

However, we recognize that the conventional classification method shown in Figure 4 based solely on the highest probability score for three classes may not achieve optimal classification performance. Thus, in order to optimize model performance, we propose a new method of classification shown in Figure 5 by adding a threshold value on the SoftMax activation function in the last layer of the trained ResNet-50 model. The idea behind increasing the threshold (0.5 to 0.9) is to identify the optimal threshold value that can yield the highest positive predictive value (PPV) in classification and minimize false positives. Specifically, after adding a threshold, a test image will only be assigned to a class in which the model-generated probability score is greater than the threshold.

After adding a threshold to SoftMax, a group of images will become undetermined if all 3 probability scores are smaller than the threshold. These undetermined images are assigned to a new class namely, Class 4, representing the “difficult” images. In order to further classify these difficult images, we add a second transfer learning model. The same pre-trained ResNet-50 model is fine-tuned again using part of these difficult images. As a result, a unique two-stage model is built. The model in the first stage is applied to identify and classify “easy” images. If the test images are classified as undetermined difficult images, they will be further analyzed and classified by another model in the second stage.

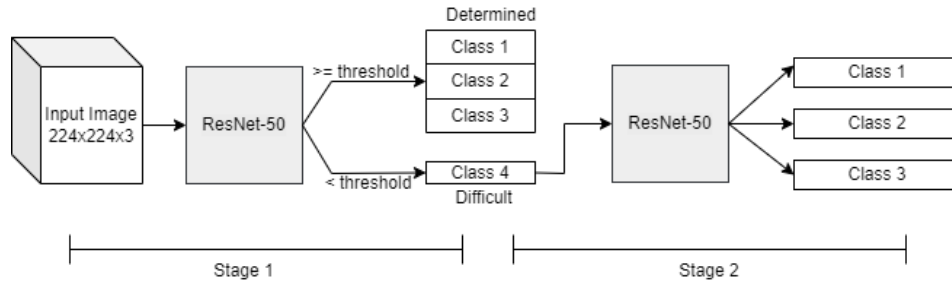


Figure 5: Block diagram of the proposed 2-stage model

Due to the limitations of the dataset size used to test the above two-stage concept, we use 5-fold cross-validation to train (fine-tune) and test the ResNet-50 model in the second stage. In both stages, the pre-trained ResNet-50 architecture is used for training. All the images are normalized and resized to $224 \times 224 \times 3$ to fit into the pre-trained ResNet-50 network. Horizontal and vertical flipping is used for Data-augmentation.

3. RESULTS

Table 1 shows performance when using one ResNet-50 model to classify images into 3 classes without adding a threshold value on the SoftMax layer. Over all 1077 test images, the model predicts and assigns 588, 109 and 380 images to Classes 1, 2, and 3, respectively. Among the classified images, true- and false-positives along with the positive predictive values are summarized in Table 1.

Table 1: Stage 1 results without threshold on the SoftMax layer

Class	1	2	3	Total
Assigned images	588	109	380	1077
True positive (TP) images	563	60	367	990
False positive (FP) images	25	49	13	87
Positive predictive value (PPV)	0.957	0.550	0.966	0.919

After adding threshold values on SoftMax layer of the model, the easy images will be classified into 3 classes and the difficult images will be assigned into Class 4 (undetermined images). As the threshold values increase from 0.5 to 0.9, the results show that PPV values of Class 1 increase, while the number of determined images decrease due to the increase of undetermined images in Class 1 (as shown in Figure 6 and Figure 7, and Table 2).

Table 2: Performance of model in Stage 1 using different threshold values on the SoftMax layer

Threshold value	0.5	0.6	0.7	0.8	0.9
Number of determined images	1034	939	799	561	234
True positive (TP) images	965	887	768	544	226
False positive (FP) images	69	52	31	17	8
Positive predictive value (PPV) of 3 Classes	0.933	0.945	0.961	0.970	0.966

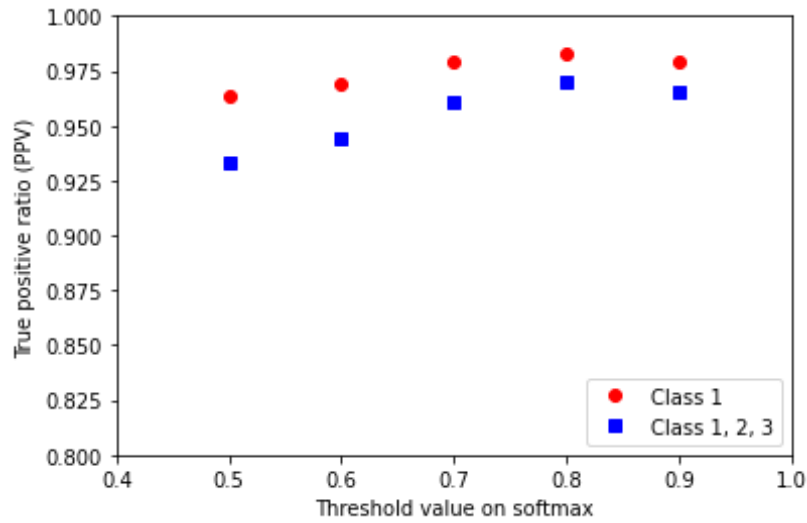


Figure 6: True positive ratio (PPV) VS Threshold value

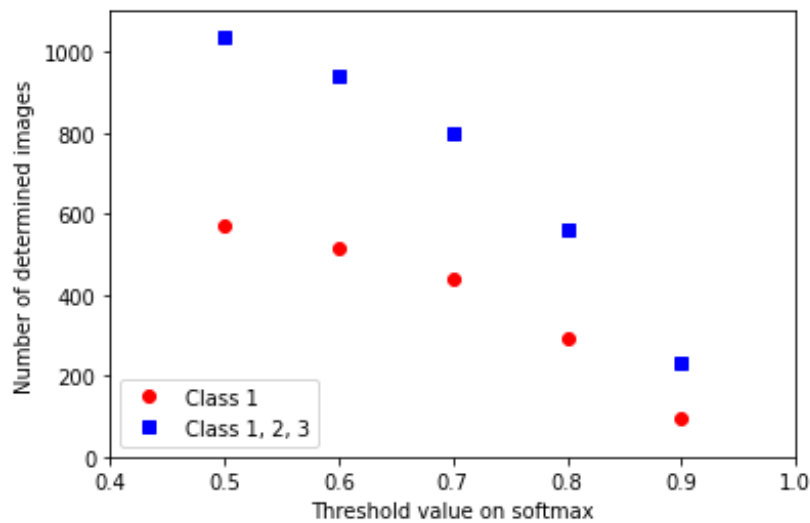


Figure 7: Number of determined images VS Threshold value

Table 3 summarizes the performance of the ResNet-50 model implemented in Stage 2 to classify the difficult images, which are undetermined by the Stage 1 model based on the SoftMax score with a threshold value of 0.8. The undetermined/difficult images ($1077 - 561 = 516$) are used in Stage 2.

Table 3: Performance of model in stage 2 with different threshold values on the SoftMax layer

Threshold value	0.5	0.6	0.7	0.8	0.9
Number of determined images	516	513	508	504	486
True positive (TP) images	490	477	489	485	476
False positive (FP) images	25	36	19	19	10
Positive predictive value (PPV) of 3 Classes	0.950	0.930	0.963	0.962	0.979

Table 4 shows the classification results using the two-stage model with threshold value = 0.8 on the SoftMax layer of the ResNet-50 model in the first stage. For example, the model in Stage 1 assigns 293 images to Class 1 and the model in Stage 2 assigns 278 images to Class 1. After running this two-stage model, a total of 571 images are assigned to Class 1. The positive predictive value (PPV) is 0.986. In comparing single-stage with multi-stage models (Table 4 and Table 1), we observe that false positives are reduced from 25 to 8 for Class 1 (the target class representing complete retinal fundus image without annotations), and from 87 to 36 for all three classes. The PPV of Class 1 increases from 95.7% to 98.6%, and the PPV over all three classes increases from 91.9% to 96.6% using two-stage model. The number of true positives for all three classes increases from 990 to 1,029 using the two-stage model. However, the total number of true positives for Class 1 remains the same using the single-stage and two-stage models.

Table 4: Classification performance using the new two-stage model with a threshold of 0.8 on the SoftMax layer of the model in the first stage.

Class	1	2	3	Total
Assigned images (Stage: 1 + 2)	293 + 278 = 571	43 + 34 = 77	225 + 192 = 417	561 + 504 = 1065
True positive (TP) images	288 + 275 = 563	33 + 19 = 52	223 + 187 = 410	544 + 485 = 1029
False positive (FP) images	5 + 3 = 8	10 + 15 = 25	2 + 5 = 7	17 + 19 = 36
Positive predictive value (PPV)	0.986	0.675	0.9832	0.966

4. DISCUSSION

In this study, we proposed, developed, and tested a new two-stage model in classifying retinal fundus images using transfer learning ResNet-50 models. Based on our experimental results, we have following key observations.

- First, using a two-stage transfer-learning model outperforms a traditional single-stage model. For instance, while observing the key statistical parameters (Table 1 and Table 4) for the entire dataset, we noticed an improvement in the PPV value from 91.9% to 96.6%. This is an increase of 4.7% as compared to the base model.
- Second, our study is primarily focused on identifying complete retinal fundus images (Class 1). On comparing the results of Class 1 between a single-stage and two-stage transfer-learning models, even though the total number of images identified are equal, a significant reduction in the false positives is noticed (from 25 to 8).
- Third, the first stage of our proposed model is essential in identifying the optimal threshold to improve the potential samples required for Stage 2. From Figure 6 and Table 2, we can see that increasing the threshold from 0.8 to 0.9 doesn't contribute toward increasing the PPV value.
- Additionally, we also observed that, like conventional machine learning or computer-aided detection schemes, performance of a deep-learning model also heavily depends on the content distribution of training datasets. Thus, it is often difficult to correctly identify or classify the difficult or subtle images using a deep-learning model that is trained or fine-tuned using a small dataset which does not contain or cannot sufficiently represent the difficult images (or outliers).
- In order to optimally address and solve this challenge, developing a two- or multi-stage deep-learning model or scheme has significant advantages. This is the primary contribution of this study to the medical imaging informatics or CAD field.

Last, we also recognize that many previous studies have highlighted the significance of integrating image processing techniques along with deep-learning frameworks [21]. Thus, in our future studies, we plan to explore this phenomenon using a more diverse dataset of retinal images and investigate application of integrated architectures that combine both traditional image processing and deep-learning frameworks.

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REFERENCES

- [1] D. L. Pham, C. Xu, and J. L. Prince, "CURRENT METHODS IN MEDICAL IMAGE SEGMENTATION 1," 2000. [Online]. Available: www.annualreviews.org.
- [2] T. McInerney and D. Terzopoulos, "Deformable models in medical image analysis: a survey," *Med. Image Anal.*, vol. 1, no. 2, pp. 91–108, Jun. 1996.
- [3] J. B. A. Maintz and M. A. Viergever, "A survey of medical image registration," *Med. Image Anal.*, vol. 2, no. 1, pp. 1–36, Mar. 1998.
- [4] C. Li, R. Huang, Z. Ding, J. C. Gatenby, D. N. Metaxas, and J. C. Gore, "A level set method for image segmentation in the presence of intensity inhomogeneities with application to MRI," *IEEE Trans. Image Process.*, vol. 20, no. 7, pp. 2007–2016, 2011.
- [5] H. Ling, S. K. Zhou, Y. Zheng, B. Georgescu, M. Suehling, and D. Comaniciu, *Hierarchical, Learning-based Automatic Liver Segmentation*. .
- [6] D. Shen, S. Moffat, S. M. Resnick, and C. Davatzikos, "Measuring Size and Shape of the Hippocampus in MR Images Using a Deformable Shape Model," *Neuroimage*, vol. 15, no. 2, pp. 422–434, Feb. 2002.
- [7] Y. Zheng *et al.*, "Automatic aorta segmentation and valve landmark detection in C-Arm CT for transcatheter aortic valve implantation," *IEEE Trans. Med. Imaging*, vol. 31, no. 12, pp. 2307–2321, 2012.
- [8] M. Betke, H. Hong, D. Thomas, C. Prince, and J. P. Ko, "Landmark detection in the chest and registration of lung surfaces with an application to nodule registration," *Med. Image Anal.*, vol. 7, no. 3, pp. 265–281, Sep. 2003.
- [9] K. Doi, "Current status and future potential of computer-aided diagnosis in medical imaging," *Br. J. Radiol.*, vol. 78, no. SPEC. ISS., pp. 21–30, 2005.
- [10] R. Bellotti *et al.*, "A CAD system for nodule detection in low-dose lung CTs based on region growing and a new active contour model," *Med. Phys.*, vol. 34, no. 12, pp. 4901–4910, 2007.
- [11] L. A. Meinel, A. H. Stolpen, K. S. Berbaum, L. L. Fajardo, and J. M. Reinhardt, "Breast MRI lesion classification: Improved performance of human readers with a backpropagation neural network computer-aided diagnosis (CAD) system," *J. Magn. Reson. Imaging*, vol. 25, no. 1, pp. 89–95, 2007.
- [12] G. Danala *et al.*, "Classification of Breast Masses Using a Computer-Aided Diagnosis Scheme of Contrast Enhanced Digital Mammograms," *Ann. Biomed. Eng.*, vol. 46, no. 9, pp. 1419–1431, 2018.
- [13] Y. Lecun, Y. Bengio, and G. Hinton, "Deep learning," *Nature*, vol. 521, no. 7553, pp. 436–444, 2015.
- [14] I. Sutskever, O. Vinyals, and Q. V. Le, "Sequence to sequence learning with neural networks," *Adv. Neural Inf. Process. Syst.*, vol. 4, no. January, pp. 3104–3112, 2014.
- [15] S. Furui, L. Deng, M. Gales, H. Ney, and K. Tokuda, "Fundamental technologies in modern speech recognition," *IEEE Signal Process. Mag.*, vol. 29, no. 6, pp. 16–17, 2012.
- [16] A. Krizhevsky, I. Sutskever, and G. E. Hinton, "ImageNet Classification with Deep Convolutional Neural Networks," in *Advances in Neural Information Processing Systems*, 2012, vol. 25, [Online].
- [17] P. Porwal *et al.*, "IDRiD: Diabetic Retinopathy – Segmentation and Grading Challenge," *Med. Image Anal.*, vol. 59, p. 101561, 2020.
- [18] M. A. Pérez, B. B. Bruce, N. J. Newman, and V. Biousse, "The use of retinal photography in nonophthalmic settings and its potential for neurology," *Neurologist*, vol. 18, no. 6, pp. 350–355, 2012.
- [19] K. He, X. Zhang, S. Ren, and J. Sun, "Deep Residual Learning for Image Recognition," *2016 IEEE Conf. Comput. Vis. Pattern Recognit.*, pp. 770–778, 2016.
- [20] K. Simonyan and A. Zisserman, "Very Deep Convolutional Networks for Large-Scale Image Recognition," Sep. 2014, Accessed: Mar. 03, 2021.
- [21] M. Heidari, S. Mirniaharikandehi, A. Z. Khuzani, G. Danala, Y. Qiu, and B. Zheng, "Improving the performance of CNN to predict the likelihood of COVID-19 using chest X-ray images with preprocessing algorithms," *Int. J. Med. Inform.*, vol. 144, p. 104284, 2020.